



CEREBRO

Your Own Personal Private AI Assistant

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Problem & Motivation

University students, new visitors, and people with visual impairments struggle daily to navigate large indoor campus environments. GPS signals are unavailable indoors, and existing tools are screen-dependent and inaccessible.

Motivation:

Make university life fun and easy for everyone — and lower stress.

Specific pain points:

- Can't find lecture room, exam room, or lab
- Can't find a specific professor / TA office or their hours
- Missed the bus — no real-time bus location or ETA
- Lectures & assignments schedule overload and confusion
- Exam preparation stress and course selection uncertainty
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CEREBRO addresses all three domains — Smart Transport, Indoor Navigation, and AI Companion — in one wearable, hands-free system with near-zero infrastructure cost.

Proposed System Design

1) High-Level Component Architecture

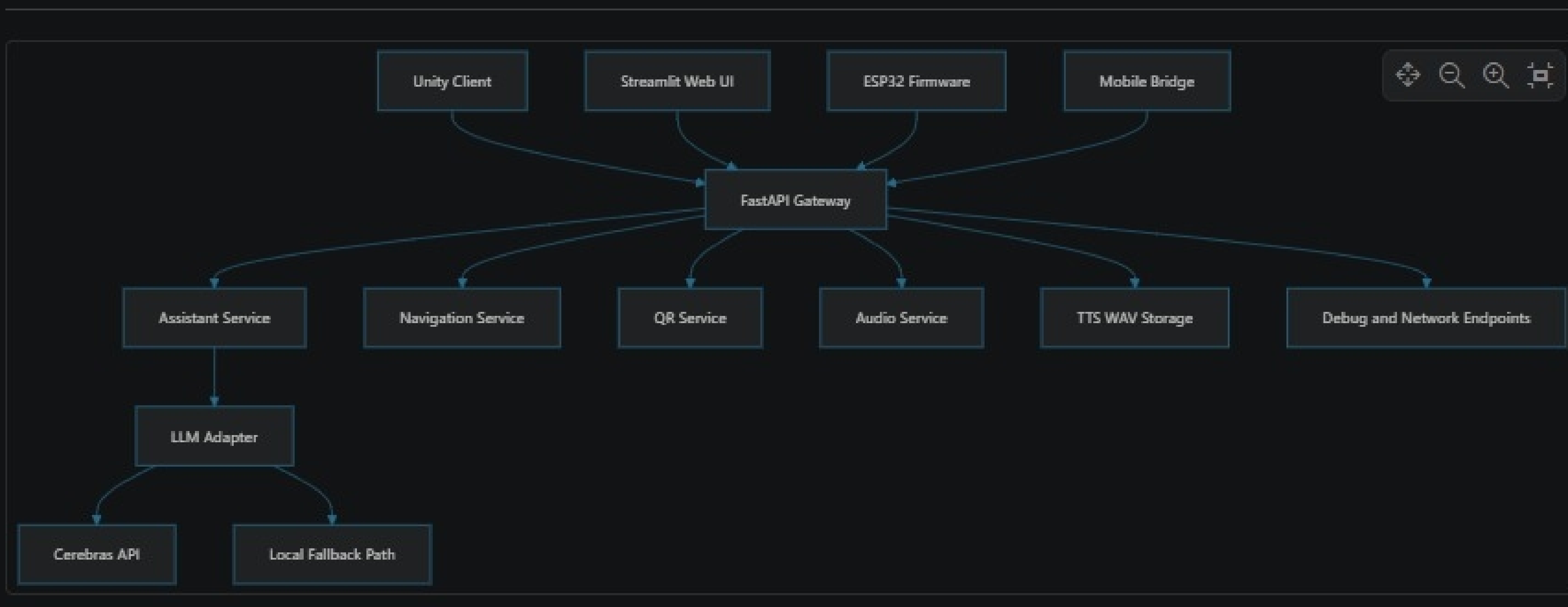


Figure 1: High-Level Component Architecture — Unity Client, ESP32 Firmware, Mobile Bridge → FastAPI Gateway → Services

2) /process Request Sequence

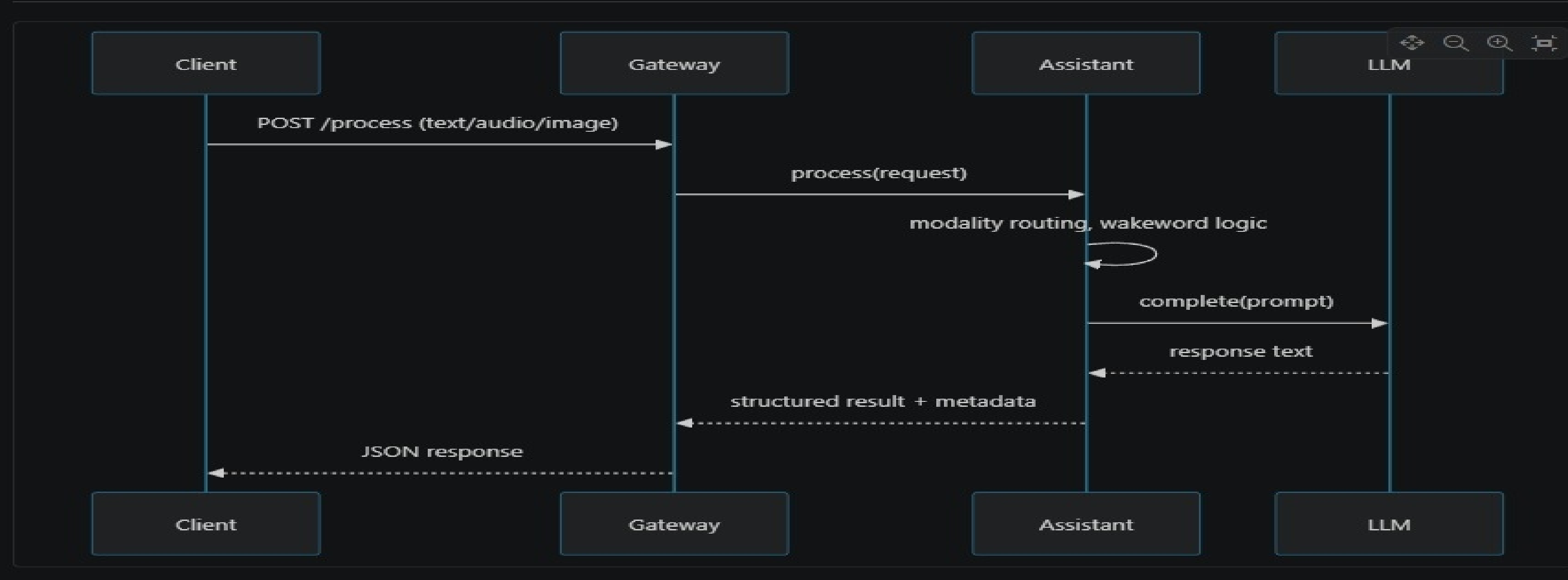


Figure 2: /process Request Sequence — Client → Gateway → Assistant → LLM → JSON response

Aim and Objectives

Build a wearable, AI-powered smart glasses system that acts as a personal private AI assistant — guiding students hands-free through campus using only voice, vision, and printed QR markers.

Core Objectives:

- ESP32-WROVER-DEV smart glasses with OV2640 camera — total BOM ~\$65.50
- Whisper STT (4.1% WER) for Arabic & English voice commands
- LLaMA 3.1-8B via Cerebras API for <600ms intent extraction
- Moondream VLM for real-time scene understanding and room identification
- A* pathfinding on 2D floorplans extracted from 3D GLB LiDAR scans
- QR-based localization achieving ±2 cm accuracy — no Wi-Fi or BLE needed
- MCP tool layer: CV, GPS, SpeechToText, TextToSpeech, NLU tools
- React Native companion app with bus tracking, indoor nav, and companion AI

Implementation and Prototype

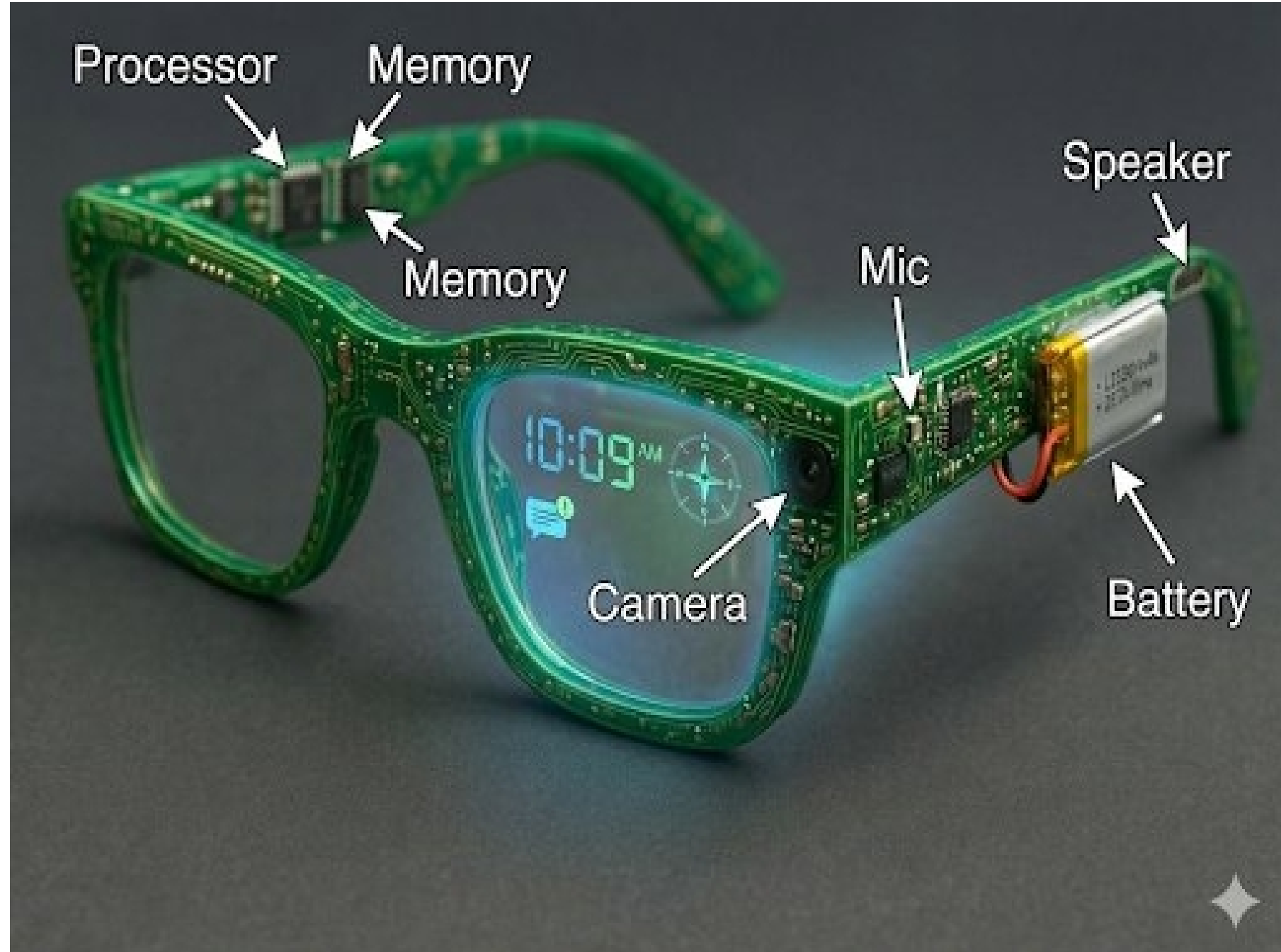


Figure 3: ESP32 Smart Glasses (Processor, Memory, Mic, Camera, Speaker, Battery)

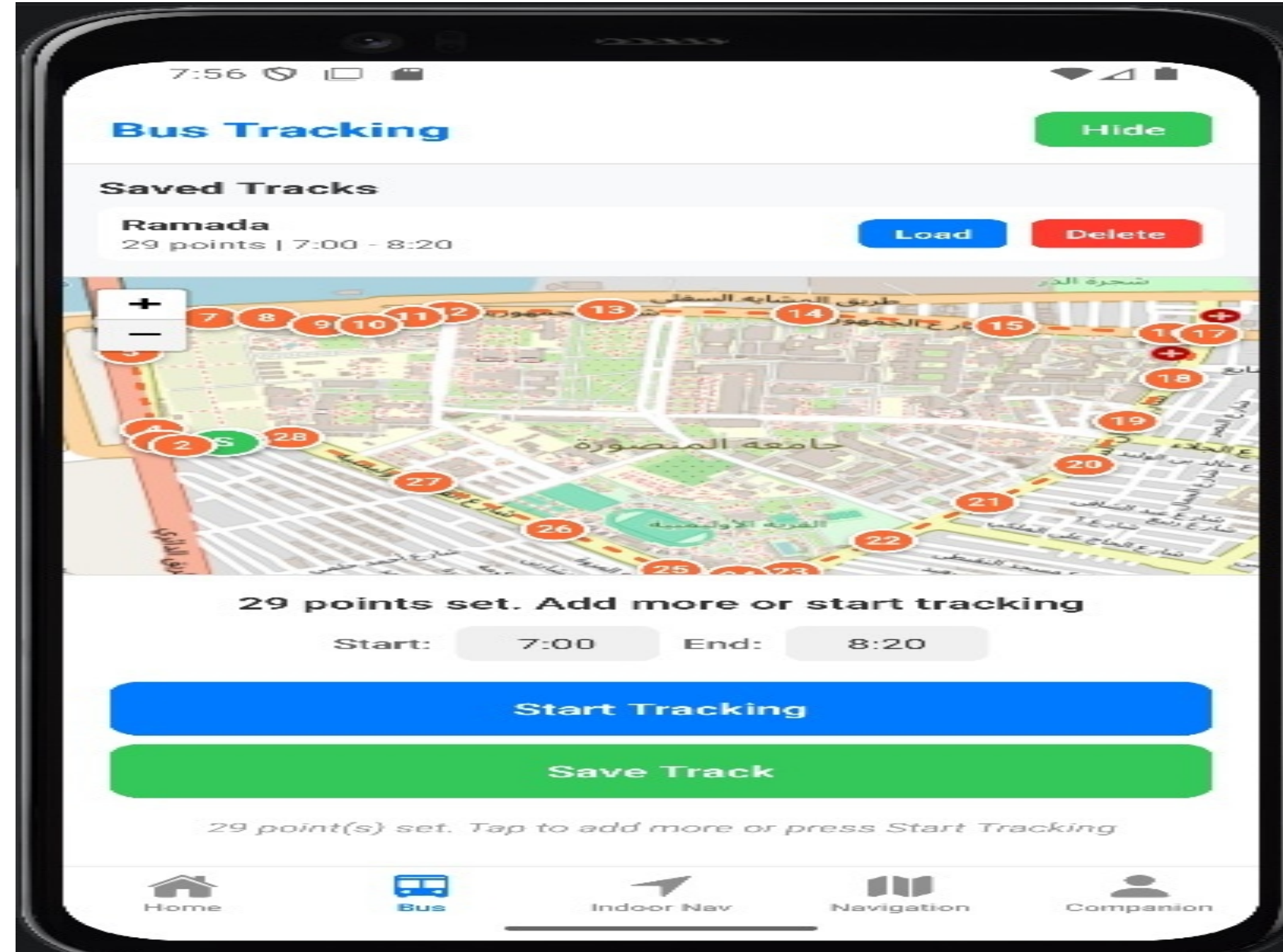


Figure 4: React Native App — Bus Tracking, Indoor Nav, Companion AI

Hardware: ESP32-WROVER-DEV + OV2640 camera, hand-etched PCB prototype, custom glasses frame. BOM cost: ~\$65.50.

AI Pipeline: Whisper STT (4.1% WER) → LLaMA 3.1-8B via Cerebras API → Moondream VLM scene analysis → A* pathfinding.

Mapping: 2D floorplan extracted from 3D GLB LiDAR scan using trimesh. QR codes for ±2 cm localization — zero infrastructure cost.

App: React Native + Three.js: Home, Bus tracking with live 29-point routes, Indoor Nav with A* runway preview, Pathverse AR, Campus finder, and Companion AI modes.

Methodology

Hardware:

- ESP32-WROVER-DEV + OV2640, hand-etched PCB, custom glasses frame
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AI Stack:

- Whisper STT — 4.1% WER, Arabic/English
- LLaMA 3.1-8B via Cerebras — 600ms intent parsing
- Moondream VLM — scene & landmark identification
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Navigation:

- GLB LiDAR scan → trimesh → 2D occupancy grid
- QR code anchors for ±2 cm localization
- A* pathfinding with step-by-step voice output
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Testing:

- Python Unit Tests + Integration Smoke Tests
- Firmware Native Tests + Unity EditMode Tests
- HIL: Unity Voice/Nav Checks + QR & ESP Checks

Results and Analysis

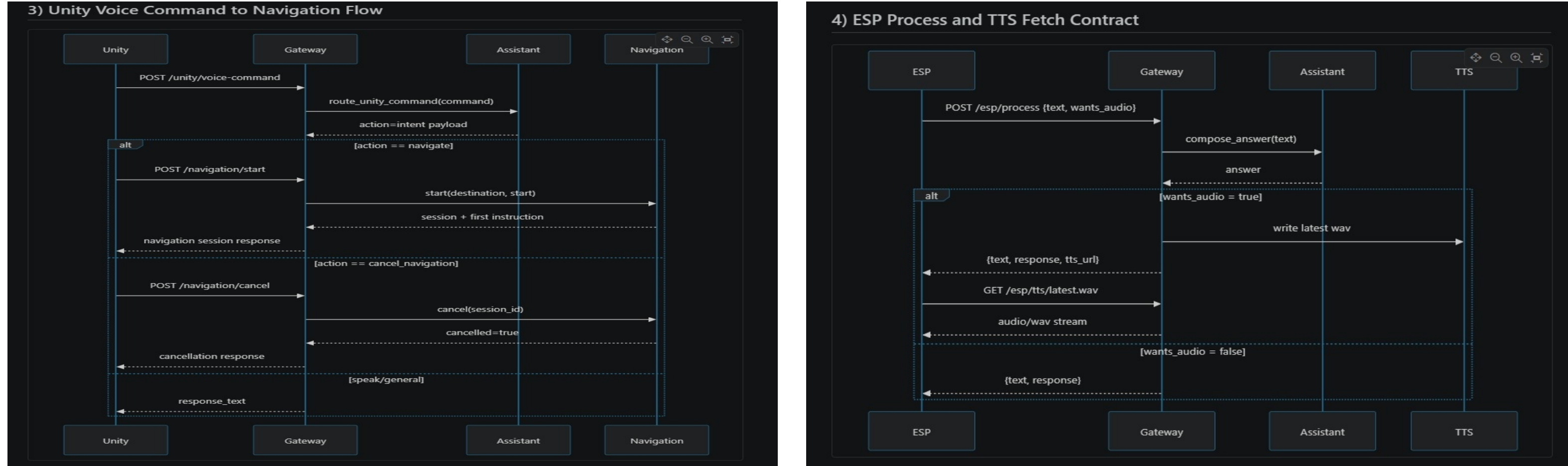


Figure 5: Voice Command to Navigation Flow (Unity→Gateway→Assistant→Navigation)

HIL 13/13 OK — All hardware-in-the-loop checks passed

E2E Latency: 1.85 s (Whisper 350ms + LLM 600ms + Nav 150ms + TTS 750ms)

Intent Precision: 94.2% (LLaMA-3.1-8B via Cerebras, 500+ requests tested)

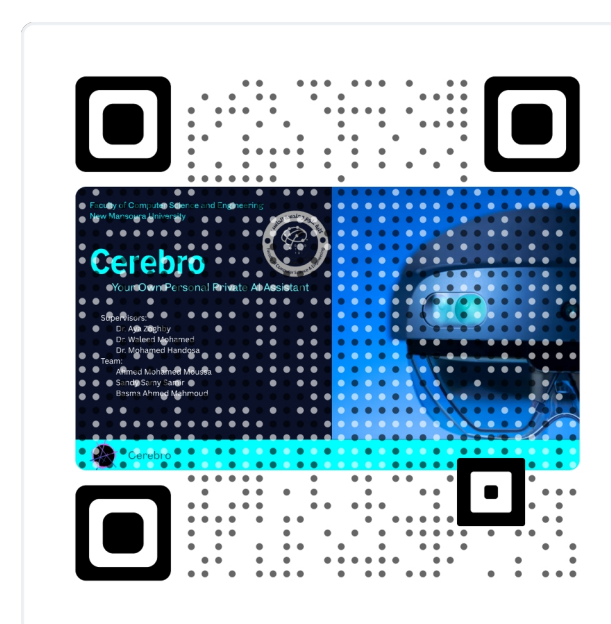
Pathfinding: 100% destination resolution on all annotated rooms

HIL Pass Rate: 98.5% across Python unit, integration, firmware, and Unity tests

QR Localization: ±2 cm accuracy — vs. Wi-Fi (±1-3 m) and GPS (unavailable indoors)

Peak Temp: 42 °C — confirms thin-client approach: no heavy AI on the glasses

QR



Scan to view demo

Conclusions and Future Developments

Conclusions

Thin-client works: ~1.85s E2E, ~42°C peak — no heavy on-glass AI required. All inference offloaded to the FastAPI gateway.

A* + QR localization works indoors: Reliable paths with ±2 cm fixes — outperforming Wi-Fi (±1-3m) and GPS (unavailable indoors) at near-zero infrastructure cost.

HIL + layered tests: 98.5% HIL pass rate across Python unit, integration, firmware native, and Unity EditMode tests — stable APIs and fewer hardware surprises.

Limitations: Ambient noise degrades STT accuracy. Cloud LLM inference requires network. Vision metrics may not cover all edge cases.

Future Developments

- Expand map library — automated digitization of more NMU buildings (hospital, library)
- Multilingual voice support and contextual follow-up queries
- YOLO object detection in AR — highlight exits, stairs, and points of interest
- Continue hardware manufacturing using CAD software; miniaturise PCB into single glasses frame
- Pilot programme with 50 students/staff — collect feedback for commercial readiness
- Expand to established AR glasses: Meta Ray-Ban and MetaQuest



Figure 7: Test & Validation Pipeline (run_all_tests.py → HIL checks → live_hil_report.json)